

Mutual linearity from classical Markov response theory

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We show that mutual linearity of stationary probabilities and currents in Markov jump processes follows directly from classical response theory. Using the exact finite-response identity expressed through the Drazin inverse, we show that changing the rates of one edge changes all stationary probabilities by the same scalar factor multiplied by fixed coefficients. Eliminating this factor gives the known mutual-linearity relations, valid arbitrarily far from equilibrium. We generalize this property beyond Markov jump processes to linear systems.

I. INTRODUCTION

Stationary probabilities and currents generally depend nonlinearly on transition rates. Nevertheless, varying the rates of one edge produces affine relations between these quantities, known as mutual linearity [1–6]. We aim to give a short derivation of mutual linearity from the classical finite-response identity [7, 8]. Using the Drazin inverse, this identity shows that all probability changes share one scalar factor. Eliminating it, we find the known relations. In addition, we formulate an extension of mutual linearity beyond Markov processes to linear systems.

II. EXACT STATIONARY RESPONSE

The finite-response identity shown below dates back to the 1960s [7] and was originally formulated for Markov chains. Here we simply rewrite it for continuous-time Markov jump processes on a finite state space.

We consider a finite irreducible Markov jump process, $\partial_t \mathbf{p} = W \mathbf{p}$, where W_{ij} is the transition rate from j to i , and $W_{jj} = -\sum_{i \neq j} W_{ij}$. The stationary distribution satisfies $W \boldsymbol{\pi} = 0$ and $\mathbf{1}^\top \boldsymbol{\pi} = 1$. For a second irreducible generator $W' = W + \Delta W$, the classical fundamental-matrix identity takes the form

$$\Delta \boldsymbol{\pi} \equiv \boldsymbol{\pi}' - \boldsymbol{\pi} = -G \Delta W \boldsymbol{\pi}', \quad (1)$$

where $G = W^D$ is the Drazin inverse [9–11]. It satisfies $GW = WG = I - \boldsymbol{\pi} \mathbf{1}^\top$, $G \boldsymbol{\pi} = 0$, and $\mathbf{1}^\top G = 0$. Indeed, subtracting the two stationary equations gives $W \Delta \boldsymbol{\pi} = -\Delta W \boldsymbol{\pi}'$. Multiplication by G and normalization give Eq. (1), without assuming that ΔW is small.

III. MUTUAL LINEARITY

A. State observables

We start with the derivation of mutual linearity for probabilities and state observables. For the second matrix W' ,

we change W_{ij} and W_{ji} , keeping all other non-diagonal rates fixed and adjusting the diagonal elements to preserve probability conservation. Then, only components i and j of the vector $\Delta W \boldsymbol{\pi}'$ can be nonzero, and they sum to zero. Equation (1) therefore gives

$$\begin{aligned} \Delta \pi_n &= (G_{nj} - G_{ni})(\Delta W_{ij} \pi'_j - \Delta W_{ji} \pi'_i) \\ &\equiv h_n \alpha, \end{aligned} \quad (2)$$

where $h_n \equiv G_{nj} - G_{ni}$ is calculated from the rates before the change and does not depend on ΔW_{ij} or ΔW_{ji} . All dependence on these rate changes enters the second factor $\alpha \equiv \Delta W_{ij} \pi'_j - \Delta W_{ji} \pi'_i$. Repeating Eq. (2) for state m and eliminating α gives

$$\Delta \pi_n = \frac{h_n}{h_m} \Delta \pi_m, \quad h_m \neq 0. \quad (3)$$

Thus, (π'_m, π'_n) lies on one straight line for every choice of the new edge rates. This property was recently discovered in Ref. [4].

More generally, for state observables $O = \sum_n O_n \pi_n$ and $Q = \sum_n Q_n \pi_n$ with fixed weights, the same argument gives

$$\Delta O = \frac{\sum_n O_n h_n}{\sum_n Q_n h_n} \Delta Q, \quad (4)$$

provided that the denominator is nonzero.

Let us now consider the linear-response limit of Eq. (2):

$$\partial_{W_{ij}} \pi_n = h_n \pi_j, \quad \partial_{W_{ji}} \pi_n = -h_n \pi_i, \quad (5)$$

so that $-\partial_{W_{ij}} \pi_n / \partial_{W_{ji}} \pi_n = \pi_j / \pi_i$ whenever $h_n \neq 0$. This recovers the response ratio of Ref. [4]. Interestingly, our dynamical fluctuation-response theory [12] gives the same ratio for dynamical responses to impulsive rate perturbations at time t , with π_j / π_i replaced by $p_j(t) / p_i(t)$. Thus, this response ratio can be obtained without invoking mutual linearity under finite rate changes.

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B. Currents

Now we turn to mutual linearity between the steady-state currents. The stationary current from l to k is $J_{kl} \equiv W_{kl}\pi_l - W_{lk}\pi_k$. For an edge whose transition rates W_{kl} and W_{lk} are kept fixed, Eq. (2) gives

$$\Delta J_{kl} = (W_{kl}h_l - W_{lk}h_k)\alpha, \quad (6)$$

where only the change in the stationary distribution contributes. On the perturbed edge, the rates themselves also change:

$$\begin{aligned} \Delta J_{ij} &= W_{ij}\Delta\pi_j - W_{ji}\Delta\pi_i + \Delta W_{ij}\pi_j' - \Delta W_{ji}\pi_i' \\ &= (1 + W_{ij}h_j - W_{ji}h_i)\alpha. \end{aligned} \quad (7)$$

Eliminating α between the currents on (i, j) and another edge (k, l) gives

$$\Delta J_{kl} = \frac{W_{kl}h_l - W_{lk}h_k}{1 + W_{ij}h_j - W_{ji}h_i} \Delta J_{ij}, \quad (8)$$

which is the mutual linearity of stationary currents [1–3]. Both current changes contain the same α , so the coefficient relating them is independent of the new edge rates, provided the denominator in Eq. (8) is nonzero.

IV. DISCUSSION AND CONCLUSIONS

The argument applies to any linear system $A\mathbf{x} = \mathbf{b}$ with nonsingular A and fixed $\mathbf{b}$. If only A_{ij} changes and A' is non-

singular, subtracting the equations and eliminating $\Delta A_{ij}x_j'$ between n and m gives

$$\begin{aligned} \Delta x_n &= -(A^{-1})_{ni} \Delta A_{ij} x_j', \\ \Delta x_n &= \frac{(A^{-1})_{ni}}{(A^{-1})_{mi}} \Delta x_m, \quad (A^{-1})_{mi} \neq 0. \end{aligned} \quad (9)$$

The same proportionality holds for fixed linear observables.

The same argument applies whenever $\Delta A\mathbf{x}' = \mathbf{u}\alpha$, with fixed $\mathbf{u}$: the perturbation enters the equations in fixed proportions, with only its scalar amplitude α changing. Then $\Delta\mathbf{x} = -A^{-1}\mathbf{u}\alpha$, so eliminating α again gives mutual linearity.

The frequency-domain extension follows from this argument using $A = sI - W$. For time-independent rates, the Laplace-transformed master equation is $(sI - W)\tilde{\mathbf{p}}(s) = \mathbf{p}(0)$. We keep the same initial distribution in both systems and define the resolvent matrix $R(s) = (sI - W)^{-1}$ for $\text{Re } s > 0$. Changing the rates of edge (i, j) and eliminating the common factor between states n and m results in

$$\begin{aligned} \Delta \tilde{p}_n &= h_n(s)(\Delta W_{ij}\tilde{p}_j' - \Delta W_{ji}\tilde{p}_i'), \\ \Delta \tilde{p}_n &= \frac{h_n(s)}{h_m(s)} \Delta \tilde{p}_m, \quad h_m(s) \neq 0, \end{aligned} \quad (10)$$

where $h_n(s) = R_{ni}(s) - R_{nj}(s)$ and all transformed probabilities are evaluated at the same s . This recovers the Laplace-domain probability relations [5, 6]. Repeating Eqs. (6)–(8) with $\tilde{\mathbf{p}}(s)$ and $h_n(s)$ gives the corresponding current relations [1]. These relations hold for the Laplace transforms. At a fixed time, varying the edge rates need not produce a straight-line relation between two probabilities or two currents [6].

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